

Lab | Warehouse Feature Comparison — Platform Evaluation Matrix

Task 1: Build the Feature Comparison Matrix

Architecture Features:

Feature	Snowflake	BigQuery	Redshift	Databricks
Storage/compute separation				
Serverless option				
Multi-cloud deployment				
Auto-scaling compute				
Auto-suspend/pause				
Storage format	Proprietary (micro-partitions)	Capacitor (proprietary)	Columnar (local)	Delta Lake (open Parquet)
Query engine	Proprietary	Dremel	Proprietary	Spark / Photon

Data Management Features:

Feature	Snowflake	BigQuery	Redshift	Databricks
Time Travel (historical queries)				
Data cloning (zero-copy)				
Native data sharing	Native (no copy)	Analytics Hub	Data sharing (limited)	Delta Sharing (open)
Semi-structured data (JSON)				
Streaming ingestion				
External tables				
Schema evolution				

Ecosystem & Integration:

Feature	Snowflake	BigQuery	Redshift	Databricks
Native Spark engine				
dbt integration				
Tableau/BI connectivity				
Python/Pandas support				
ML capabilities	Snowpark (Python/Scala)	BigQuery ML (SQL)	SageMaker integration	MLflow, Spark ML
CI/CD & version control				
REST API				

Task 2: Scenario-Based Evaluation

Use Case 1: Nightly ETL Pipeline

- Process 2TB of raw events nightly using Spark
- Transform and load into curated tables
- Must complete within a 3-hour window

Platform	Score (1-10)	Justification
Snowflake	6	
BigQuery	6	
Redshift	6	
Databricks	10	Native Spark

Use Case 2: Self-Service BI Dashboards

- 20 analysts running Tableau dashboards
- Peak usage 9 AM–12 PM
- Need sub-10-second query response times

Platform	Score (1-10)	Justification
Snowflake	10	No ETL/BI interference, separate warehouses
BigQuery	8	
Redshift	8	
Databricks	6	BI remains secondary

Use Case 3: Data Sharing with Partners

- Share curated datasets with 15 retail partners
- Partners should query directly (no data copying)
- Different partners on different clouds

Platform	Score (1-10)	Justification
Snowflake	10	native, no copy
BigQuery	5	Analytics Hub
Redshift	6	Data sharing (limited)
Databricks	8	delta sharing, strong

Use Case 4: Ad-Hoc Data Exploration

- Data scientists running complex queries on historical data
- Unpredictable workload patterns

- Need Python/notebook support

Platform	Score (1-10)	Justification
Snowflake	7	
BigQuery	10	Workloads are ad-hoc / unpredictable. Automatic scale to query size
Redshift	6	less flexible
Databricks	8	Spark

Use Case 5: ML Feature Engineering

- Build feature tables from event data for ML models
- Requires Spark for complex transformations
- Need integration with MLflow or equivalent

Platform	Score (1-10)	Justification
Snowflake	6	
BigQuery	7	BigQuery ML is strong for SQL-based ML workflows
Redshift	5	Depends on external AWS ML tooling
Databricks	10	MLflow and Spark

Task 3: Weighted Evaluation

Use Case	Weight (%)
Nightly ETL Pipeline	30%
Self-Service BI Dashboards	25%
Data Sharing with Partners	20%
Ad-Hoc Data Exploration	10%
ML Feature Engineering	15%
Total	100

Platform	ETL (wxs)	BI (wxs)	Sharing (wxs)	Ad-Hoc (wxs)	ML (wxs)	Total
Snowflake	1,8	2,5	2	0,7	0,9	7,9
BigQuery	1,8	2	1	1	1,05	6,85
Redshift	1,8	2	1,2	0,6	0,75	6,35
Databricks	3	1,5	1,6	0,8	1,5	8,4

Task 4: Risk Analysis

Platform	Risk 1	Risk 2	Risk 3
Snowflake	High processing cost	Low ML integration	Spark not supported
BigQuery	Pay per query, unpredictable cost	GCP only	Less control over compute resources
Redshift	Requires more tuning	AWS only	Less advanced ML/data sharing
Databricks	Higher platform complexity	SQL experience less polished	BI less mature

Task 5: Write the Executive Summary

Databricks is the best choice as it aligns directly with the company's Spark-based engineering workflows, ML training and growing data lake architecture.

Key Strengths of Databricks

- Native Spark architecture ideal ETL pipeline
- Strong ML ecosystem with MLflow
- Multi-cloud
- Open lakehouse architecture using Delta Lake on existing S3 storage